

Servo Tuner

Bench Tool User Guide

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KJO Liquid Rocket Motor Control System

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A standalone bench tool for finding the PWM open/close endpoints of a servo-actuated ball valve, using a Serial console to move the servo and read back its position potentiometer.

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1 Purpose

Servo Tuner is a small standalone sketch used on the bench to determine the correct PWM_OPEN/PWM_CLOSE calibration constants for a single servo-actuated ball valve, before those values are hardcoded into production firmware (EMU, GSEMU). It drives one servo channel and reads one position potentiometer, letting the operator move to a candidate PWM count, read the resulting pot value, and iteratively nudge the target until the valve sits exactly at its mechanical open or closed stop.

This is a different program from the **EMU Servo Tuner** (a related but separate tool that tunes all three of EMU's installed valves against its actual per-valve pin/channel table – see that tool's own guide).

2 Hardware Required

- Adafruit Feather M0 (ATSAMD21G18A)
- Adafruit 8-channel PWM/Servo FeatherWing (PCA9685), I²C
- One servo-actuated ball valve, servo wired to **PCA9685 channel 7**
- The valve's position potentiometer wired to **analog pin A1**
- USB cable for Serial (115200 baud)

Note

The servo channel (7) and pot pin (A1) are hardcoded in the sketch, not configurable at runtime. To tune a valve wired to different pins, edit `servo_channel/pot_channel` in `main.cpp` and reflash.

3 How to Use

Open a Serial monitor at 115200 baud after powering the board. The tool runs an outer “new target” loop and an inner “adjust” loop:

1. **Target:** the tool prints a suggested PWM range (see §3.1) and prompts for a target PWM count. Enter a number and press Enter.
2. **Hysteresis delta:** enter a count value. Every subsequent move first drives the servo `target + delta` counts past the target, then back to the target itself – this removes mechanical backlash from the reading, so the pot value reflects the valve consistently approaching from one direction.
3. The servo moves to the target and the tool prints the resulting potentiometer reading (`Encoder = ...`).
4. **Action prompt** – type one letter and press Enter:
 - **a** (adjust): enter a signed delta (e.g. +20 or -15); the tool adds it to the current target, re-approaches through the hysteresis move, and prints the new pot reading.
 - any other key: returns to step 1 for a completely fresh target and hysteresis delta.

Repeat the adjust step, watching the printed pot value, until it matches the valve's known fully-open or fully-closed pot reading – the PWM count at that point is the calibration value to hardcode into production firmware.

3.1 Suggested Target Range

The range printed at each “New Target” prompt is computed at compile time from a generic 5-turn-servo, 60:16-gear, 90°-travel ball valve (the same geometry used across the production valves) – it is a starting-point hint, not a hard limit, since PWM counts still need individual per-valve trimming on the bench:

$$\text{start} \approx 2329 \quad \text{end} \approx 3918$$

These numbers come from the servo’s 1100 Hz PWM frequency, its 500–2500 μs mechanical range, the 60:16 gear ratio, and 90° of required valve travel – see `main.cpp`’s `constexpr` derivation if you need to recompute them for a different gear ratio or travel angle.

4 Notes

- The Serial console has no timeout – the tool waits indefinitely at every prompt.
- Backspace is supported while typing a value.
- There is no “undo” – if a bad target damages the mechanism, power-cycle and re-enter values carefully.

5 Version History

Version	Date	Notes
0.1	September 2026	Initial user guide.
